

CLASS XII · LECTURE 14

Microsoft Access and SQL

Both options · Two periods · Section C: data types in MS Access. Objects, queries, one SELECT

By the end of this lecture you can

- Name the Access objects and the job of each.
- List the field data types with one example each.
- Write a SELECT statement with a WHERE clause.
- Distinguish selecting rows from projecting columns.

The objects on the Access window

Object	Job	You use it when
Table	Stores the data	You design Student with RollNo, Name and Marks
Query	Asks a question or changes rows	You want only the students who passed
Form	A screen for viewing and typing	A clerk enters a new admission
Report	A layout for paper	You print a class list
Macro	A saved sequence of actions	A button opens a form
Module	Visual Basic code inside Access	A calculation the macro cannot express

The object used to display and print data, in the paper's wording, is the **report**. The object that stores the data is the **table**. A query does not store the data permanently; it shows a view of it. Do not answer "query" when the question says "print".

Data types

A data type is the kind of value a field may hold. Choosing it badly wastes space or rejects legal values. This list is the Section C answer. Newer Access renames Text to Short Text and Memo to Long Text. If you have learned the new names, write the old name and the new name together. Markers taught on either screen can follow you.

Data type	Holds	Example field
Text (Short Text)	Words and short codes, up to 255 characters. Digits that are not quantities, such as a phone number	Name, RollNo stored as text
Memo (Long Text)	A long note	Remarks
Number	A quantity you will add or compare	Marks, Age
Date/Time	A date, a time, or both	AdmissionDate
Currency	Money, without binary rounding surprises	Fee
AutoNumber	A number Access fills in, different for every new row	A receipt id
Yes/No	A true or false	FeePaid
OLE Object	A file embedded from elsewhere, such as a picture	Photo
Hyperlink	A link or a web address	Portfolio URL

Use Text for a roll number if you will not add the roll numbers up. Use Number for marks because you will. Use Date/Time for a date rather than Text, so that "after March" is a real comparison. AutoNumber is not typed by the user. Yes/No is not the place for a student's name.

SQL that you can write by hand

SQL is the language of the relational model. In this paper you need to read and write a basic **SELECT**. You should also recognise **INSERT**, **UPDATE** and **DELETE** as the statements that add, change and remove rows.

```
SELECT Name, Marks
FROM Student
WHERE Marks >= 40;
```

*Query 14.1 Names and marks of students who passed. The asterisk form **SELECT *** would return every column.*

Projection chooses columns: the **Name**, **Marks** list. **Selection** chooses rows: the **WHERE** clause. A question that says “differentiate selection and projection” is this pair. Selection is horizontal (which records). Projection is vertical (which fields).

```
CREATE TABLE Student (
    RollNo TEXT(10),
    Name TEXT(40),
    Marks NUMBER
);

INSERT INTO Student (RollNo, Name, Marks)
VALUES ('1101', 'Ali', 81);
```

Query 14.2 A new table and one inserted row. Types are written so the example matches the lecture.

In the Access design grid you can do the same query without typing SQL: choose the table, tick the columns, and put **>= 40** in the Criteria row under Marks. Knowing both the grid and the **SELECT** is the practical. The written paper wants the words and the statement.

A **join** brings columns from two tables together. An equi-join keeps the rows where the keys are equal, for example Student.DeptCode equal to Department.DeptCode, so each student appears with the department name. You can describe that in a sentence if a short question says “join”. A full syntax of every join type is beyond what this paper usually demands.

EXAM TIP The long answer on data types

Open with one sentence: a data type decides what a field may store and what operations are sensible. Then eight types, each with a field name from a Student or Fee table. Close with the roll-number point: text if you will not calculate, number if you will. That is a complete 10-mark answer.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. In MS Access, the object used to print data is the:

- A) Table
- B) Report
- C) Module
- D) Macro

2. A column in a table is also called:

- A) A tuple
- B) An attribute or field
- C) A form

- D) A macro

Short answers

S1. Differentiate selection and projection.

S2. Write a query to show Name and Marks of students who scored 80 or more.

Detailed answer

L1. Explain the data types used in MS Access.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. A table stores. A report is laid out for paper.

MCQ 2. B. A tuple is a row. An attribute is a column.

S1. Selection picks rows. In SQL it is the WHERE clause, for example WHERE Marks >= 40. Projection picks columns. It is the list after SELECT, for example SELECT Name, Marks. Selection is which records. Projection is which fields.

S2. SELECT Name, Marks FROM Student WHERE Marks >= 80; Name and Marks are the projection. The WHERE clause is the selection.

L1. A data type specifies the kind of value a field can store. Text (Short Text) stores names and codes up to 255 characters; use it for a name or a roll number you will not add up. Memo (Long Text) stores a long remark. Number stores quantities such as marks or age. Date/Time stores a date or a time, such as the day of admission. Currency stores money, such as a fee. AutoNumber gives each new row a unique number automatically. Yes/No stores a true or false, such as whether the fee is paid. OLE Object stores an embedded file such as a photograph. Hyperlink stores a web address. Choosing Number for marks allows a total. Choosing Text for a phone number stops Access from dropping a leading zero.